

## **Project Title: SecureAuthX: A Three-Layer Authentication System**

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### **Abstract:**

**SecureAuthX** is a three-layer authentication system developed to provide enhanced security for web applications by combining **password authentication, OTP verification, and facial recognition**. Traditional password-based login systems are vulnerable to attacks such as phishing, brute-force attacks, and credential theft. To address these challenges, SecureAuthX implements a multi-factor authentication approach that verifies a user's identity through three sequential security layers.

The authentication process begins with username and password verification, followed by a time-sensitive One-Time Password (OTP) sent to the user's registered email. After successful OTP validation, the system performs facial recognition by comparing the user's live facial image with the stored facial data to confirm identity. Passwords are securely encrypted before storage, ensuring sensitive user information remains protected.

By integrating knowledge-based, possession-based, and biometric authentication, SecureAuthX significantly improves account security while maintaining a simple and user-friendly login experience. The system is suitable for applications that require strong user authentication, such as banking, healthcare, educational institutions, and enterprise portals.

### **Technologies Used**

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Java, Spring Boot
- **Database:** MySQL
- **Face Recognition:** OpenCV
- **Authentication:** Email OTP (JavaMail API / SMTP)
- **Development Tools:** IntelliJ IDEA / Eclipse, Maven
- **Version Control:** Git, GitHub